

ANTENATAL CARE STANDARD

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1. Standard Scope and Purpose

1.1 Purpose

- 1.1.1 This standard sets out the service specifications and minimum requirements for Healthcare Providers for provision of pre-conception and antenatal care in the Emirate of Abu Dhabi aimed at reducing maternal and infant mortality rates and enhancing the overall quality of care.
- 1.1.2 This standard will apply to all DoH licensed healthcare facilities and professionals delivering antenatal health services in the Emirate of Abu Dhabi. This encompasses private, governmental, and semi-governmental hospitals.

1.2 Scope

- 1.2.1 This standard is intended to ensure that all pregnant women receive the requisite screening and counseling during the pre-conception and antenatal periods. To do this, it mandates:
 - 1.2.1.1 A screening schedule for pre-conception and antenatal screening.
 - 1.2.1.2 A recommended education and counseling program during the pre-conception and antenatal periods
 - 1.2.1.3 Data recording and reporting requirements for antenatal care.
 - 1.2.1.4 An outline for the conditions considered as high risk for which a pregnant woman should be referred to an Obstetric physician for ongoing care.

2. Definitions and Abbreviations

No.	Term / Abbreviation	Definition
2.1	Antenatal Care	The care administered by proficient healthcare practitioners to pregnant women and adolescent girls, aiming to optimize the health of both mother and baby throughout pregnancy.
2.2	Pre-conception Care	The delivery of biomedical, behavioral, and societal health measures to women and couples before conception, with the goal of enhancing reproductive health outcomes. It generally includes screening for risks, health promotion and education, and interventions to address identified risks, all taking place before a woman attempts to become pregnant until she becomes pregnant.
2.3	Electronic Medical Record (EMR)	A digital version of a patient's record, which contains the medical and treatment history of the patients.
2.4	Fetal Growth Restriction (FGR)	A condition in which a fetus does not achieve its genetically predetermined potential size, often indicated by an estimated weight below the 10th percentile for its gestational age.

3. Standard Requirements and Specifications

3.1 Service Specifications:

The provider must:

- 3.2.1.1** Provide and support pre-conception and antenatal services as detailed in Appendix 1 periodically or as determined necessary by the licensed medical practitioner responsible for the patient's care.
- 3.2.1.2** Comply with the DoH guidelines for patient consent and ensure that individuals receive appropriate and adequate information relevant to the service and grant consent prior to screening. The consent process, including any refusals to screenings, must all be documented.
- 3.2.1.3** Ensure that when offering any assessment, intervention or procedure, the risks, benefits, and implications are discussed with the patient.
- 3.2.1.4** A family-centered approach to preconception and antenatal care should be taken, considering the patient / family's views, beliefs, values, and rights, and facilitating them in making informed decisions about their care.
- 3.2.1.5** Preconception and antenatal care cover a wide range of services, including pre-pregnancy counselling and education, requires screenings and structured antenatal care.
- 3.2.1.6** Ensure that there is effective and prompt communication between healthcare professionals of different specialties who are involved in the woman's care during pregnancy.

3.3 Preconception Care:

- 3.3.1.1** Since most pregnancies are unplanned, preconception health should be assessed with all patients of childbearing age during routine check-ups.
- 3.3.1.2** Preconception care includes health promotion and education, screening for risks, and interventions to address identified or potential risks.
- 3.3.1.3** Health promotion and/or intervention includes pregnancy planning, physical examination, cervical cancer screening (if not done previously), vaccination (if indicated), careful management of acute /chronic conditions (e.g. diabetes or hypertension) and healthy lifestyle (nutrition, exercise, and avoidance of teratogenic agents like alcohol, tobacco, Isoretinoin etc.).
- 3.3.1.4** Healthcare providers should initiate a conversation about family planning and fertility as part of comprehensive education.
- 3.3.1.5** Comprehensive counseling and education to be provided regarding the benefits of supplementation before and during pregnancy on maternal and infant health. Supplementation includes folic acid, iron, vitamin C (to improve iron absorption), and docosahexaenoic acid (DHA); it is preferred to prescribe a single multi-vitamin to increase patient compliance.

3.4 Antenatal Care

All antenatal care should be clearly documented in the electronic medical record. Risk assessment should be done at booking and at every antenatal visit thereafter (Refer to Appendix 1- Routine Antenatal Screening and Care).

3.4.1 Routine Antenatal Care

- 3.4.1.1** The first booking should be early in pregnancy, ideally by 10 weeks. Assessment includes detailed history, confirmation of pregnancy, physical examination and counselling/education.
- 3.4.1.2** Beyond the first visit, the medical professional should conduct the visits as outlined in Appendix 1

- 3.4.1.3** Review the woman's previous medical records including records held by other healthcare providers if available (check Malaffi Health Information Exchange or ask the patient for documents).
- 3.4.1.4** If there are no contraindications or risk factors in the woman's medical, surgical, and obstetric history, she will be designated as low risk and can have antenatal care in the primary care setting. If identified as high-risk (refer Appendix 2), care should be provided according to the Tiered Maternity Framework.
- 3.4.1.5** The standard makes recommendations on baseline clinical care for all pregnant women. Pregnant women with high-risk conditions (Refer to Appendix 2) usually require additional care to be determined by treating physician based on their clinical judgment.
- 3.4.1.6** Ask the woman about her general health and wellbeing (e.g. sleep, nutrition, physical activity, emotional support).
- 3.4.1.7** Ask the woman (and her partner, if present) if there are any concerns they would like to discuss.
- 3.4.1.8** Provide a safe environment and opportunities for the woman to discuss topics such as concerns about the birth (for example, if she previously had a traumatic birth) or mental health concerns. All discussions should support shared decision making and be tailored to the woman's needs, preferences, and stage of pregnancy.
- 3.4.1.9** Review and reassess the plan of care for the pregnancy.
- 3.4.1.10** Assess risk for fetal abnormality as per the provisions in Appendix 3
- 3.4.1.11** Assess risk for fetal growth restriction (Refer to DoH Antenatal Ultrasound Guideline- Appendix 5)
- 3.4.1.12** If the woman has had any hospital admission or significant health event since her last appointment, assess her risk of venous thromboembolism.
- 3.4.1.13** Midwives should be involved and referred to across the entire journey based on the roles and responsibilities outlined in the Midwife Scope of Practice (2023)
- 3.4.1.14** Pregnant women shall be offered antenatal education during consultation visits. Education should include management of labor, pain management, breastfeeding education, postnatal care, nutrition, and importance of newborn screening.
- 3.4.1.15** Update the woman's antenatal records to include details of the latest additional history, test results, examination findings, medicines, and discussions.
- 3.4.1.16** In primary healthcare setting:
 - 3.4.1.16.1** At each antenatal visit, assess for risk factors as per high-risk pregnancy criteria and refer for care to an Obstetric physician if meets criteria (refer to Appendix 2)
 - 3.4.1.16.2** All low-risk pregnancies to have their care transferred to an Obstetric physician between 28 and 32-weeks' gestation or at any gestation after 32 weeks for women who present late.
- 3.4.1.17** All pregnant women with criteria for high-risk pregnancy (as per Appendix 2) must be referred to the Obstetric physician led care from the booking visit.
 - 3.4.1.17.1** A clear plan of care must be documented in the woman's medical record.
 - 3.4.1.17.2** After the obstetric consultation, the obstetrician will advise the woman and family physician of the plan of care and if the woman remains suitable for low-risk care or needs to transfer to obstetrician led care.

3.4.2 High-risk Antenatal Care

- 3.4.2.1** If a patient is identified as high-risk as per the provisions in Appendix 2, care should be provided according to the Tiered Care Framework.
- 3.4.2.2** A clear referral path as established in 'Standard for Center of Excellence in High-risk Pregnancy and Neonates' should be followed so that highly critical cases are treated by the appropriate team of Consultant Obstetricians and other relevant specialties.

4. Key Stakeholder Roles and Responsibilities

4.1. Licensed Healthcare Professionals:

All DoH licensed and authorized health professionals involved in antenatal care must:

- 4.1.1. Maintain their competencies and satisfy DoH requirements for continuing medical education and professional development.
- 4.1.2. All healthcare professionals should hold an active DoH license as per the Professionals Qualification Requirements (PQR) and work within their scope of practice.
- 4.1.3. Abnormality screenings as outlined in Appendix 3, including antenatal screening for fetal abnormality, aneuploidy screening and non-invasive perinatal test, should be conducted by a qualified healthcare professional trained in gynecological and obstetrics ultrasound procedures as outlined in Obstetrics and Gynecology Scope of Practice, with specific focus on detection of congenital anomalies.
- 4.1.4. Obstetric and family medicine physicians in a primary care setting will ensure that physicians / midwives who deliver preconception and antenatal care have received necessary training and maintain current competencies.
- 4.1.5. Comply with the DoH Standard for Clinical Privileging and Scopes of Practice.
- 4.1.6. Provide clinical services in accordance with the requirements of this Standard, and the relevant DoH Standards. Provide the latest and the best possible screening and treatments available for antenatal conditions.

4.2. Licensed Healthcare Providers

- 4.2.1. Antenatal care services shall be provided in healthcare facilities that are licensed by DoH to provide these services. The healthcare facility should meet the HF requirement as per the DoH Health Facility Guidelines.
- 4.2.2. Ensure that the multi-disciplinary teams must comprise the necessary personnel and staff with requisite qualifications and skills (e.g., Obstetricians/Gynecologists (OB/GYNs), midwives, neonatologists, nurses, anesthesiologists, nutritionist).
- 4.2.3. Comply with the Ministerial Resolution No. (14) of 2021 on the Patient's Rights & Responsibilities Charter and deliver culturally and socially relevant patient information and education.
- 4.2.4. Comply with DoH policies and standards on managing patient medical records, including developing and maintaining the patient records, the records confidentiality, privacy and security of patient information.
- 4.2.5. Report and submit e-Claims and other healthcare data to comply with approved DoH standards regulating data sharing, medical records, patient privacy, etc.
- 4.2.6. Document and monitor quality and safety of clinical care and outcomes of antenatal patients and make these available to DoH for auditing, as and when requested to do so.

4.3. Payers / Third-Party Administrators:

- 4.3.1. Payment will follow current reimbursement methodologies in accordance with Standard Provider Contract, DoH Mandatory Tariff and associated Claims and Adjudication Rules (all available on the DoH Shafafiya webpage).
- 4.3.2. Payment will remain consistent with patient's insurance policy and schedule of benefits.
- 4.3.3. Health Insurers are responsible for ensuring the requirements set out in this Standard and relevant UAE laws and regulations are met.

5. Monitoring and Evaluation

A monitoring and evaluation framework is in place to evaluate the effectiveness, outcomes, and impact of this Standard, and where necessary adopt changes to ensure continuous improvement within the health system in line with emerging new developments in healthcare sciences, medical practices, healthcare education and training.

Continuous monitoring will be undertaken to review the operations across the facilities and to identify any emerging issues. In addition to this, audits will be conducted to review the performance of the facilities and the compliance to the standard, as deemed necessary.

A list of KPIs is outlined in Appendix 4 to this Standard for facilities to continuously track and monitor. However, the respective facilities should monitor any further metrics as they deem fit based on their clinical judgement.

6. Enforcement and Sanctions

- 6.1.** Healthcare providers, payers and third-party administrators must comply with the terms and requirements of this Standard, the DoH Standard Contract and the DoH Data Standards and Procedures.
- 6.2.** DoH may impose sanctions in relation to any breach of requirements under this Standard in accordance with the disciplinary regulations of the healthcare sector.

7. Related Documents (Appendices)

- 1. Routine Antenatal Screening and Care
- 2. Criteria for High-risk Pregnancies
- 3. Antenatal Screening for Fetal Abnormality
- 4. Key Performance Indicators (KPIs)

8. Relevant Reference Documents

No.	Reference Date	Reference Name	Relation Publication Links
1	August 2021	NICE guideline: Antenatal care	www.nice.org.uk/guidance/ng201
2	February 2013	RCOG Green-top Guideline No. 31 The Investigation and Management of the Small-for-Gestational-Age Fetus	https://www.rcog.org.uk/media/t3lmi/hnl/gtg_31.pdf
3	September 2020	ISUOG Practice Guidelines: diagnosis and management of small-for-gestational-age fetus and fetal growth restriction	https://www.isuog.org/static/b2aa3fb4-031e-4d84-b7246d613a466884/ISUOG-Practice-Guidelines-diagnosis-and-management-of-small-for-gestational-age-fetus-and-fetal-growth-restriction.pdf
4	July 2022	DoH, Antenatal Ultrasound Guideline	https://www.doh.gov.ae/-/media/B9EB96B7586B4C329F0C95DC15A624CB.ashx
5	2021	Joint Commission International Accreditation Standards for Hospitals	https://store.jointcommissioninternational.org/jci-accreditation-standards-for-hospitals-7th-edition/
6	December 2023	DoH, Midwife Scope of Practice	https://www.doh.gov.ae/-/media/9E4FA6DFE88C4E809B796392BB0C1971.ashx
7	December 2023	DoH, Standard for Center of Excellence in High-risk Pregnancy and Neonates	https://www.doh.gov.ae/-/media/11950FF9BC9E4A108D925EB454773FB1.ashx
8	March 2023	DoH, Standard for Clinical Privileging	https://www.doh.gov.ae/-/media/87DBE2A3226940E8A5712D3577064672.ashx
9	2021	Ministerial Resolution No. (14) of 2021 on the Patient's Rights & Responsibilities Charter	https://www.ehs.gov.ae/App_Content/Legislations/EHS-LAW-EN-1/files/basic-html/page1.html
10	September 2020	DoH, Standard on Patient Healthcare Data Privacy	https://www.doh.gov.ae/-/media/Feature/Aamen/DOH-Standard-on-Patient-Healthcare-Data-Privacy.ashx
11	January 2016	HAAD Guidelines for Patient Consent	https://www.doh.gov.ae/-/media/2034E1E3C99C4C4F9E4A96133575439D.ashx

Appendix 1: Routine Antenatal Screening and Care

Gestation	Screening	Counselling and Education	Interventions
Preconception Visits	Patients History: - Inquire about the patient's past weight loss efforts or bariatric surgery, and current medications, encompassing procedures, but not limited to, Laparoscopic Band, Roux en Y Gastric Bypass, Sleeve Gastrectomy, and weight loss drugs such as Ozempic, Wegovy, Semaglutide, Saxenda, and other GLP-1 agonists. - Inquire about history of mental illness. Investigations: - Medical and family history - Risk assessment - Height, weight, BMI, blood pressure - Physical examination - Rubella, varicella, vitamin D screening Cervical smear if indicated. - Rh status - Pap smear -	- Nutrition and weight - Physical activity - Smoking cessation - List of medications - Benefits of vitamins and supplements in pre-conception period such as but not limited to Folic Acid, Iron, Vitamin C, Vitamin D, DHA, and Calcium. - Accurate recording of menstrual dates - Importance of early prenatal care - Importance of continuity of care throughout the pregnancy journey - Importance of preventative dental visit - Genetic counselling** - Discuss risks of weight management procedures and medications in pregnancy, and disclosure to the OBGYN/ Midwife - Control of chronic diseases -	- Rubella vaccination if indicated. - Varicella Vaccination if indicated. - Hepatitis B vaccination - Prenatal multi-vitamin (PNV) supplement containing all critical vitamins (e.g., Folic Acid, Iron, DHA) - Additional supplementation requirement based on facts of the case. - In case of people with history of weight management procedures, continually monitor nutrient levels to detect any deficiencies
1st Antenatal Visit (Ideally by 10 weeks)	History taking: - Full obstetric history including any complications for each previous pregnancy. - If previous pregnancy $\geq 24+0$ weeks gestation at delivery, include gestation, mode of delivery, gender and birth weight of baby. - Medical, gynecology, surgical and family history - Review previous medical records including records from other healthcare providers. - Review current medicines and allergies. - Risk assessment for pregnancy complications such as diabetes, pre-eclampsia, venous thromboembolism - Risk assessment for fetal growth restriction - Identify women with risk factors who made need additional care. - Inquire about the patient's past weight loss efforts or bariatric	- Physical activity - Nutrition and diet - Discuss risks of weight management procedures and medications in pregnancy, and disclosure to the OBGYN/ Midwife - Benefits of vitamins and supplements in pre-conception period such as but not limited to Folic Acid, Iron, Vitamin C, Vitamin D, DHA, and Calcium - Common physiological conditions in pregnancy. - o Nausea and vomiting (multiple small meals, intake of ginger, Vitamin B6, Doxylamine) - o Constipation/ hemorrhoids (increasing dietary fiber and water intake, considering laxatives) - o Heartburn (having small frequent meals, relevant medications) - o Low back/ Pelvic pain (exercises, physiotherapy, support belts)	- PNV supplement containing all critical vitamins (e.g., Folic Acid, Iron, DHA) - Additional supplementation requirement based on facts of the case. - If opts for aneuploidy screening book appointment for this at appropriate time (e.g. NT at 11-14 weeks gestation) - In case of people with history of weight management procedures, continually monitor nutrient levels to detect any deficiencies

Gestation	Screening	Counselling and Education	Interventions
	surgery, and current medications, encompassing procedures, but not limited to, Laparoscopic Band, Roux en Y Gastric Bypass, Sleeve Gastrectomy, and weight loss drugs such as Ozempic, Wegovy, Semaglutide, Saxenda, and other GLP-1 agonists. - Lifestyle history including occupation, living condition, smoking and alcohol intake. Physical examination - Height, weight (including 1st trimester weight), BMI. - BP - Assessment of gestational age (See DOH Antenatal Ultrasound Guidelines: Appendix 2) Investigations: - Booking bloods including rubella susceptibility, HIV, syphilis, Hepatitis B surface antigen - Varicella: if no history of chicken pox - Vitamin D: if indicated. - CBC - Hemoglobinopathy, if not done previously - Sickle cell screening: ethnicity from high MCH > 27 - ABO/Rhesus/Antibody screen - Urine dipstick and urine culture sensitivity - RBG, FBG, hBA1C	- o Leg cramps (Magnesium, calcium supplements) - o Varicose veins (compression socks, leg elevation) - Importance of continuity of care throughout the pregnancy journey - Discuss antenatal screening, including fetal anatomy scan. - Discuss screening for fetal aneuploidy. - Importance of preventative dental visit - Describe Home Visit Program and secure consent	
11 – 14 weeks	- Weight, BP and urine dipstick - Review NT scan results and number of fetuses - Correct and finalize EDD (see DOH Antenatal Ultrasound Guidelines; Appendix 2) - Discuss and offer NIPT test if indicated or requested. - Inform patient of result, discuss and document care plan for the pregnancy. - Risk assess for pregnancy complications and refer for additional care if indicated	- Breastfeeding Counselling - Nausea and vomiting - Reiterate benefits of vitamins and supplements during pregnancy	- Advise those at risk of PET or FGR to start 150 mg aspirin once daily at night. - Book follow up appointment at 16 – 18 weeks
16 - 18 weeks	- Weight, BP and urine dipstick - Listen to fetal heart. - Risk assess for pregnancy complications and refer for additional care if indicated	- Nutrition and weight - Reiterate benefits of vitamins and supplements during pregnancy - Breastfeeding	- Book anatomy scan for 18+0 – 22+6 weeks gestation - If indicated refer to fetal medicine services (See DOH Antenatal Ultrasound

Gestation	Screening	Counselling and Education	Interventions
	- Explain and offer detailed anatomy scan at next appointment. - If maternal Rh negative, conduct fetus Rh through Fetal cell-free DNA test		Guidelines: Appendix 6) for uterine artery Doppler to be done at 22 – 24 weeks
19 - 23 weeks	- Weight, BP and urine dipstick - Review results of anomaly scan - Listen to the fetal heart if no scan is done at the same appointment. - Risk assess for pregnancy complications and refer for additional care if indicated - Discuss GTT and its importance	- Healthy lifestyle - Nutrition and exercise - Reiterate benefits of vitamins and supplements during pregnancy - Smoking cessation - GDM	- Book OGTT appointment for 24 -28 weeks gestation - Book growth scan for 28 weeks if indicated by FGR risk assessment (See DOH Antenatal Ultrasound Guidelines: Appendix 5)
25 weeks	- Weight, BP and urine dipstick - Listen to the fetal heart if no scan is done at that appointment. - Risk assess for pregnancy complications and refer for additional care if indicated - Discuss the baby's movements. - Advise her to contact maternity services at any time of day or night if she has concerns about her baby's movements or noticed reduced fetal movements. Discuss with the woman her birth preferences, their implications and their benefits and risks	- Health lifestyle - Nutrition and exercise - Fetal growth - Awareness of fetal movement - Antenatal classes - GDM - Breastfeeding - Continuity of care - Reiterate benefits of vitamins and supplements during pregnancy	- Book OGTT appointment for 24 – 28 weeks gestation - Book growth scan for 28 weeks if indicated by FGR risk assessment (See DOH Antenatal Ultrasound Guidelines: Appendix 5) - Pre-order CBC and antibody screen tests to be done at 28 weeks
28 weeks	- Weight, BP and urine dipstick - Review growth scan if done and plot on a fetal growth chart. - If no growth scan, listen to fetal heart. - Measure SFH in women with a singleton pregnancy unless the woman is having regular growth scans or SFH has been measured less than 2 weeks ago. - Review GTT results and discuss implications if GDM diagnosed. - CBC and antibody screen - Discuss benefits and offer TDAP vaccination. - Discuss fetal movements, and when and where to report if any concerns. - Risk assesses for pregnancy complications and refer for additional care if indicated	- Healthy lifestyle - Nutrition and exercise - Awareness of fetal movement - Gestational diabetes management and care plan if indicated. - Discuss birth plan (incl. mode of delivery, labor, induction preferences) - Preparing for labor and birth - Recognizing active labor - Postnatal period includes caring for the newborn and breastfeeding, Vitamin K prophylaxis. - Advise the woman to avoid going to sleep on her back after 28 weeks pregnancy.	- If SFH < 10th centile, book growth scan in 1 week. - If EFW < 10th centile or AC < 5th centile, refer to fetal medicine services (See DOH Antenatal Ultrasound Guidelines: Appendix 6) - Review OGTT results and refer to dietician and diabetes nurse if GDM diagnosed. - RSV and TDAP vaccine - CBC and antibody screen - If mother Rh negative and fetal blood group positive, or not known, offer Anti D prophylaxis - If fetal cell free DNA show the fetus is Rh negative, there is no need to offer Anti D

Gestation	Screening	Counselling and Education	Interventions
	- For low-risk women routine growth scan not to be done from 34 weeks gestation - For low-risk women SFH to be measured every 4 weeks from 26/28 weeks till delivery - For high-risk women growth scans to be done from 24/26 weeks every 2-4 weeks till delivery and ideally plotted on a growth chart (See DOH Antenatal Ultrasound Guidelines: Appendix 5)		
32 weeks	- Weight, BP and urine dipstick - Review growth scan if done and plot on a fetal growth chart. - If no growth scan, listen to fetal heart. - Measure SFH in women with a singleton pregnancy unless the woman is having regular growth scans or SFH has been measured less than 2 weeks ago. - Review GTT results and discuss implications if GDM diagnosed. - CBC and antibody screen if not done at 28 weeks. - Discuss fetal movements, and when and where to report if any concerns. - Risk assess for pregnancy complications and refer for additional care if indicated - For low-risk women routine growth scan not to be done from 34 weeks gestation - For low-risk women SFH to be measured every 4 weeks from 26/28 weeks till delivery - For high-risk women growth scans to be done from 24/26 weeks every 2-4 weeks till delivery and plotted on a growth chart (See DOH Antenatal Ultrasound Guidelines: Appendix 5)		- Book routine third trimester growth scan for 34-36 weeks - If SFH < 10th centile book growth scan to be done within 1 week - If EFW < 10th centile or AC < 5th centile, refer to fetal medicine services (See DOH Antenatal Ultrasound Guidelines: Appendix 6) - Manage pregnancy complications as per relevant guidelines. - For primary care, transfer antenatal care to hospital care for the rest of the pregnancy preferably to the hospital the woman intends to deliver her baby
34 weeks	- Weight, BP and urine dipstick - Review growth scan if done and ideally plot on a growth chart. - If no growth scan listen to fetal heart - Measure SFH in women with a singleton pregnancy unless the woman is having regular growth	- Healthy lifestyle - Nutrition and exercise - Review birth plan including signs of labor, explain signs of pregnancy complications like pre-eclampsia, obstetric cholestasis.	- Book growth scan if indicated. - If SFH < 10th centile book growth scan within 1 week - If EFW < 10th centile or AC < 5th centile, refer to fetal medicine services (See DOH Antenatal Ultrasound Guidelines: Appendix 6)

Gestation	Screening	Counselling and Education	Interventions
	scans or SFH has been measured less than 2 weeks ago. - Discuss fetal movements and when and where to report if any concerns. - Risk assess for pregnancy complications and refer for additional care if indicated	- Discuss GBS screening and reasons why its offered to be done ≥ 35 weeks	- Manage pregnancy complications as per relevant guidelines
36 weeks	- Weight, BP and urine dipstick - Review growth scan if done and plot on-a growth chart. - If no growth scan listens to fetal heart - Measure SFH in women with a singleton pregnancy unless the woman is having regular growth scans or SFH has been measured less than 2 weeks ago. - Discuss fetal movements and when and where to report if any concerns. - Risk assess for pregnancy complications and refer for additional care if indicated - Ask the woman and her partner if they have any other concerns they would like to discuss. - Culture for Group B Strep (≥ 35 weeks)	- Healthy lifestyle - Nutrition and exercise - Review birth plan including signs of labor, pain management, options for labor. - Explain signs of pregnancy complications like pre-eclampsia and obstetric cholestasis	- Obtain culture for Group B Strep - CBC and antibody screen - Book growth scan if indicated. - If SFH <10 th centile book growth scan within 1 week. - If EFW <10 th centile or AC < 5 th centile, refer to fetal medicine services (See DOH Antenatal Ultrasound Guidelines: Appendix 6) to be seen in 1 week. - Manage pregnancy complications as per relevant guidelines. - Book weekly antenatal review appointments from 37 - 40 weeks gestation

Key:

BMI:	Body mass index	HIV:	Human immunodeficiency virus
BP:	Blood pressure	IOL:	Induction of labor
CS	Cesarean section	MCH	Mean corpuscular hemoglobin
ECV:	External cephalic version	NIPT	Non-invasive prenatal tests
EDD:	Estimated date of delivery	NT:	Nuchal translucency
EFW:	Estimated fetal weight	OB:	Obstetric
FBG:	Fasting blood glucose	OGTT:	Oral glucose tolerance test
FGR:	Fetal growth restriction	PET:	Pre-eclampsia
GBS:	Group B Streptococcus	Rh:	Rhesus
GDM:	Gestational diabetes mellitus	RBG:	Random blood glucose
HbA1c	Hemoglobin A1c	SFH:	Symphysial fundal height
PNV	Prenatal plus multivitamin		

* If clinically indicated ** If not done previously

Appendix 2: Criteria for High-risk Pregnancies

Pregnant women with any of the following conditions usually require additional joint care with Consultant Obstetrician and relevant specialties.	Women who have experienced any of the following in a previous pregnancy usually require additional joint care with Consultant Obstetrician and relevant specialties.
- Cardiac disease - Hypertension - Renal disease - Endocrine disorder or diabetes requiring insulin / or metformin, GLP1 receptor Agonist drugs. - Psychiatric disorders (being treated with medication) - Haematological disorder - Autoimmune disorder - Epilepsy requiring anticonvulsant drugs. - Malignant disease - Severe asthma - HIV or Hepatitis B infection - Obesity (BMI 35 or above) or underweight (BMI ≤ 18) - Pre-labour, preterm rupture of membrane (PPROM) - Higher risk of developing complications for example, women aged 40 and older, women who smoke. - Women who are particularly vulnerable (such as teenagers) or who lack social support. - Previous weight loss surgery such as bariatric surgery, procedures, lap band, sleeve gastrectomy, RYGB, etc. - Abnormal screening tests / fetal abnormality - Spinal, pelvis or back trauma / injury - Multiple Sclerosis/Neurological disorder - Rheumatoid/Systemic Lupus Erythematosus Disease - Liver Disease	- Recurrent miscarriage (three or more in first trimester miscarriages or 1 second trimester loss) - Preterm birth (birth < 37 weeks) - Severe pre-eclampsia, Eclampsia, Hemolysis, Elevated Liver Enzymes and Low Platelets (HELLP) syndrome or eclampsia. - Rhesus isoimmunisation or other significant blood group antibodies - Uterine surgery including known Uterine isthmocoele (uterine niche), caesarean section, myomectomy or cone biopsy. - Antenatal or postpartum haemorrhage on two occasions - Puerperal psychosis - Grand multiparity of 6 or more pregnancies - A stillbirth or neonatal death - A small for gestational-age infant (birthweight below 10th centile) - A large for gestational-age infant (birthweight above 95th centile) - A baby with birthweight below 2.5 kg or above 4.5 kg - A baby with a congenital abnormality (structural, genetic or chromosomal). - Obstetric cholestasis - History of Cervical incompetence/ cerclage or cone biopsy/ LEEP procedure - Confirmed placenta accreta. - Threatened premature labor (fibrinectin positive) or PPRM <30 weeks. - Anticipated anesthesia-related problems (e.g., previous failed intubation, anesthetic) - Hyperthermia and severe needle phobia

Appendix 3: Antenatal Screening for Fetal Abnormality

1. Antenatal screening for fetal abnormality

- All women should have a discussion at booking visit (1st and 2nd trimester) about screening for fetal abnormality.
- They should be told about ultrasound screening and its limitations.
- They should be informed about aneuploidy screening tests and options i.e. nuchal translucency (NT) scan (if they are less than 14 weeks) and non-invasive perinatal testing (NIPT)

2. Aneuploidy screening

- Offer first trimester aneuploidy screening to all women (low and high risk (nuchal translucency (NT) or combined NT test between 11 weeks to 14 weeks (or CRL 45 to 84 mm)
- Offer non-invasive perinatal testing (NIPT) for high-risk cases (see below)
- Offer second trimester screening test or NIPT if booked late for first trimester screening.
- Documentation to reflect the discussion of the offered test and if the patient declined the test.

3. Ultrasound Screening (see DOH Antenatal Ultrasound Guidelines, July 2022)

- Second trimester anatomy/ detail scan is routinely offered to all patients between 18 - 23+6 weeks gestation (ideally planned for 20+ weeks); high risk patients will have 2nd trimester detailed anomaly scan in fetal medicine services as per referral criteria (See DOH Antenatal Ultrasound Guidelines: Appendix 6).
- If the patient has had the 2nd trimester scan outside the hospital, you do not need to repeat the anomaly scan, but you should review the report, with documentation in the note to reflect the above.
- If >24 weeks and has not had anomaly scan elsewhere order Baseline Scan (CPT 76805) and ask for anomaly review in scan order but explain to patient that late anomaly scan may be difficult to complete.

4. List of high-risk women who should always be offered non-invasive perinatal testing (NIPT)

- Maternal age ≥ 35
- High nuchal translucency (NT), i.e. NT > 95th centile (should be referred to fetal medicine services if NT ≥ 3 mm)
- Abnormal first trimester combined NT screening test (BHCG & PAPP-A) / second trimester serum screening
- Previous history of aneuploidy (these women should be referred to fetal medicine services for discussion & consultation).
- Fetal abnormality if decline invasive prenatal diagnostic testing.

Appendix 4: Key Performance Indicators

No.	KPI	Definition	Numerator	Denominator	Unit of Measure	Desired Direction
01	Obstetric History Completion Rate	Percentage of women who have had their comprehensive obstetric history recorded during their first antenatal visit.	Women with completed obstetric history at 1st check-up	Women registered for 1st antenatal check-up	Percentage	Higher is better
02	Early Booking Rate for Care and Delivery	Percentage of women who registered for both pregnancy care and delivery services at the facility before 20 weeks of gestation.	Women booked for care and delivery at <20 weeks	Total pregnant women	Percentage	Higher is better
03	Anomaly Scan Coverage Rate	Percentage of women who received an anomaly scan between 18 to 24 weeks of gestation.	Women with anomaly scan at 18-24 weeks	Women registered for antenatal care	Percentage	Higher is better
04	Comprehensive Hemoglobin Screening Rate	Percentage of women who received Hemoglobin checks at booking, 28 weeks, and 36 weeks.	Women with Hb checked at specified weeks	Women registered for antenatal care	Percentage	Higher is better
05	Preeclampsia Screening Rate	Percentage of women screened for hypertensive disorders like preeclampsia after 20 weeks of gestation.	Women screened for hypertensive disorders	Women registered for antenatal care after 20 weeks	Percentage	Higher is better
06	Supplementation Counselling Rate	Percentage of women who were counselled on and prescribed essential supplementation.	Women counselled and prescribed supplements	Women registered for antenatal care	Percentage	Higher is better
07	Early to Mid-Pregnancy Engagement Rate	Average number of antenatal care visits attended by women during weeks 4-28.	Total number of visits by women during weeks 4-28	Distinct count of number of women who had visits during weeks 4-28	Number	Higher is better
08	Late Pregnancy Engagement Rate	Average number of antenatal care visits attended by women during weeks 28-36.	Total number of visits by women during weeks 28-36	Distinct count of number of women who had visits during weeks 28-36	Number	Higher is better
09	Pre-Delivery Engagement Rate	Average number of antenatal care visits attended by women during weeks 36-40.	Total number of visits by women during weeks 36-40	Distinct count of number of women who had visits during weeks 36-40	Number	Higher is better
10	Timeliness of First Antenatal Visit	Percentage of women categorized by gestational age at their first antenatal visit: <11 weeks.	Number of women having their first visit at <11 weeks	Women registered for antenatal care	Percentage	Lower is better
11	Delivery Planning Rate	Percentage of women who have booked their delivery at	Women delivered by week 20	Total delivered women	Percentage	Higher is better

		the facility by 20 weeks gestation.				
12	Birth Plan Establishment Rate	Percentage of women who have created a birth plan by 36 weeks gestation.	Women with a birth plan by week 36	Women registered for antenatal care more than 36 week	Percentage	Higher is better
13	GBS Screening Adoption Rate	Percentage of women who received GBS screening.	Women with GBS screening before delivery	Women registered for antenatal care more than 35 weeks	Percentage	Higher is better
14	Growth Monitoring Rate	Percentage of women for whom a growth chart was generated during antenatal visits.	Total number of women for whom a growth chart was generated during antenatal visits	Women registered for antenatal care more than 28 weeks	Percentage	Higher is better
15	SFH Plotting Consistency Rate	Percentage of women who had SFH measured and plotted more than twice.	Women with SFH plotted >2 times	Women registered for antenatal care more than 28 weeks	Percentage	Higher is better
16	High-Risk Fetal Monitoring Rate	Percentage of high-risk women who had EFW measured and plotted more than twice.	High-risk women with EFW plotted >2 times	High-risk women registered for antenatal care more than 28 weeks	Percentage	Higher is better